A high-confidence causal bQTL atlas reveals the regulatory architecture of natural gene expression variation in maize

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Abstract

Binding quantitative trait loci (bQTL) identify sequence variants that alter transcription factor (TF) occupancy, but linking individual variants to causal gene expression changes remains challenging due to linkage disequilibrium and motif redundancy. Here, we apply a four-step filtering pipeline to 147,942 maize bQTL from a 25-hybrid pan-cistrome to isolate 1,126 high-confidence *causal* variants under well-watered conditions and 1,267 under drought stress—positions where a single nucleotide change at a single bQTL per gene causally shifts allele-specific expression (ASE) across multiple NAM hybrids (FDR < 0.05). Only 12% of causal positions overlap between conditions, yet 81% of target genes are condition-specific.

We demonstrate that 41.1% of variants simultaneously disrupt one TF family's motif and create another—with zero self-switches—revealing that causal bQTL *rewire* TF occupancy rather than simply ablate it. Per-hybrid motif linkage identifies driver motifs at 92.7% of causal bQTL across 33 TF families. A directed TF regulatory network (6,787 edges) reveals that 83 transcription factor genes are themselves causal bQTL targets, creating amplification cascades.

Across 24 hybrids, the total regulatory variant burden is remarkably uniform (CV $= 0.04$), while the specific genes targeted differ substantially (Jaccard $= 0.304$)—a "distributed regulatory operating system" where individual variants drift but the aggregate grammar is conserved by stabilizing selection. Multi-bQTL effects are predominantly additive (2.6:1 over saturating), causal variants are depleted in transposable elements (12.2% vs 20.6%), and 73.7% show concordant binding-to-expression causal chains. Under drought, WRKY disruption increases 34%, transport genes are enriched 1.6-fold, and aquaporin TIP3-2 emerges as the strongest drought-causal target.

This atlas provides a prioritized set of regulatory switches suitable for functional validation, precision breeding, and as training data for sequence-based regulatory models.

1 Introduction

A central question in quantitative genetics is how sequence variation at cis-regulatory elements translates into phenotypic diversity. In maize, the pan-cistrome study by Engelhorn et al. [Engelhorn et al., 2025] identified 147,942 binding quantitative trait loci (bQTL)—positions where at least one of 25 F1 hybrids (B73 $\times$ diverse founders) shows allelic differences in TF binding relative to the B73 reference. These bQTL collectively explained the majority of heritable trait variation for over 70% of 143 phenotypes tested, establishing cis-regulatory variation as a dominant force in maize biology.

However, not every bQTL is equally informative. Many bQTL are in linkage disequilibrium (LD), cluster in the same promoter, or disrupt motifs that have redundant copies nearby. To move from correlative association to mechanistic understanding, we need to identify variants where we can confidently attribute a causal chain: *this specific nucleotide* disrupts *this specific TF binding site*, which shifts expression of *this specific gene*.

Here, we describe a systematic filtering approach that distills 147,942 bQTL down to 1,126 high-confidence causal variants under well-watered conditions and 1,267 under drought stress. Each passes four criteria: (1) it is the *only* significant bQTL within 3 kb of its target gene's TSS, eliminating LD ambiguity; (2) the genotype at this position causally predicts allele-specific expression (ASE) across 24 genotyped NAM hybrids (FDR < 0.05); (3) it is the single bQTL per gene, ensuring unambiguous causal attribution; and (4) the 3 kb distance threshold is data-driven, corresponding to where the causality rate drops to the 5% FDR baseline. We then characterize these causal variants across multiple analytical dimensions to reconstruct the regulatory architecture of natural expression variation.

2 Results

2.1 A four-step pipeline identifies 1,126 causal bQTL

We developed a filtering pipeline (Table 1) that progressively enriches for variants with unambiguous causal interpretation. Starting from 147,942 bQTL positions with allelic binding differences, we mapped each to gene promoters within 3 kb of the transcription start site (TSS), retaining bQTL–gene pairs with available allele-specific expression (ASE) data across 24 genotyped NAM hybrids.

The critical step is the genotype-stratified causal test. For each bQTL–gene pair, we split the 24 genotyped NAM hybrids into two groups based on their allele at the bQTL position, then compared the magnitude of allele-specific expression between groups using a Mann–Whitney U test. P-values were corrected using the Benjamini–Hochberg procedure. The 3 kb distance threshold was determined empirically as the distance at which the causality rate (fraction of bQTL passing FDR < 0.05) drops to the 5% baseline, ensuring we capture the distance regime where bQTL are informative about target gene expression.

To avoid LD confounding, we restricted to genes with exactly one significant bQTL in their promoter, yielding the final set of 1,126 genes with high-confidence causal bQTL.

Table 1: **Filtering pipeline from bQTL to causal variants.**

Step	Filter	N remaining	Drop
1	bQTL positions (allelic binding differences)	147,942	Non-variable
2	bQTL within 3 kb of gene TSS + ASE data	19,368 pairs	Distal or no ASE
3	Genotype $\rightarrow$ ASE causal test, FDR < 0.05	1,403 pairs	Non-significant
4	Single bQTL per gene (no LD ambiguity)	1,126 genes	Multi-bQTL

2.2 Linkage disequilibrium and the single-bQTL design

Among the 1,403 significant bQTL–gene pairs, the majority of genes have exactly one significant bQTL, presenting no linkage concern. For the remaining genes with multiple bQTL, 62% of within-gene variant pairs have identical genotype vectors across the 24 founders (complete LD), making it impossible to distinguish which variant is causal. Our single-bQTL filter removes this ambiguity at the cost of excluding some real signals. This conservative design ensures that every gene in the 1,126 set has an unambiguous causal variant.

2.3 Within binding peaks, disrupted motifs show family-specific redundancy

A key concern with single-nucleotide regulatory variants is apparent redundancy: if a promoter contains multiple copies of a TF motif, disrupting one should have little effect. We addressed this by scanning the MOA-seq binding peak sequence for each of the 1,126 bQTL against all 405 DAP-seq PWMs (33 TF families). For each bQTL, we identified the tightest peak containing the variant across all 25 hybrids, then counted copies of each disrupted TF family's motif elsewhere within that peak.

Of 1,126 bQTL, 1,079 (95.8%) fall within at least one hybrid's binding peak. Among these, 880 (78.2%) have at least one identifiable TF family motif at the variant position. **31.0%** (349/1,126) have at least one disrupted family where the variant-overlapping motif is the sole copy of that family within the peak, and 24.8% (279/1,126) have all disrupted families as sole copy.

Sole-copy rates vary dramatically across TF families (Figure 1). Families with longer, information-rich motifs—HSF (31.9%), GATA (13.3%), bZIP (11.5%)—show high sole-copy rates within the ∼160 bp peak. In contrast, families with short, degenerate motifs—MYB (0.3%), C2H2 (0.4%), MYB-related (1.0%)—maintain multiple copies even within narrow peaks. This family-level variation correlates with motif sensitivity vs buffering: high-sole-copy families ("sensitive") show positive correlation between disruption and ASE effect size, while low-sole-copy families ("buffered") show negative or no correlation.

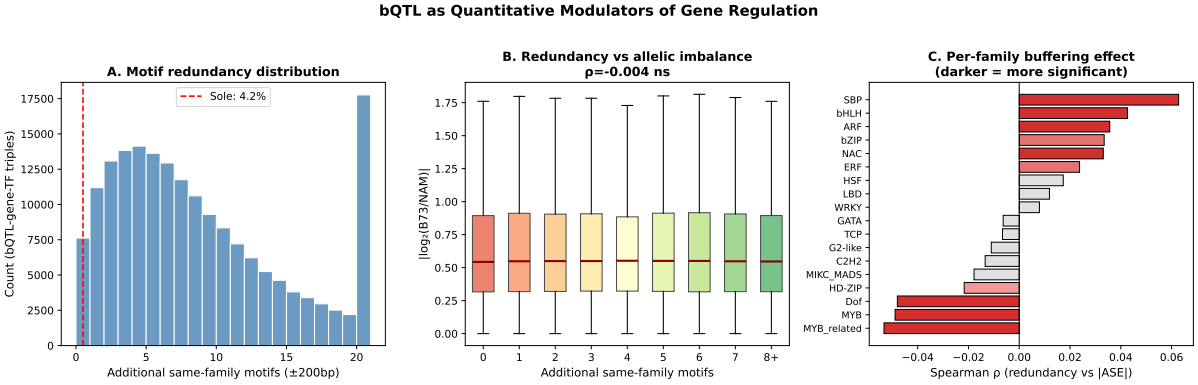


Figure 1: **Motif redundancy within MOA-seq binding peaks.** Sole-copy rates vary by TF family, with sensitive families (HSF, bZIP) showing high sole-copy rates and buffered families (MYB, C2H2) maintaining multiple copies within ∼160 bp peaks.

2.4 Multi-family disruption follows a billboard model

57.6% of causal bQTL simultaneously disrupt motifs from two or more TF families at the variant position (Figure 2). We tested whether multi-family disruption produces stronger expression effects. Strikingly, there is no dose-response relationship between the number of disrupted families and Cohen's d (Spearman $\rho = 0.049$, $p = 0.16$). Regression models predicting effect size from disrupted family indicators yield $R^2 < 0$ across all specifications, indicating that TF family identity has no predictive value for the magnitude of allelic expression differences.

This result supports the *billboard model* of cis-regulatory evolution [Arnosti and Kulkarni, 2003]: individual TF binding sites contribute independently rather than synergistically, and the primary determinant of variant impact is whether the variant is in open chromatin, not which specific TF families it affects.

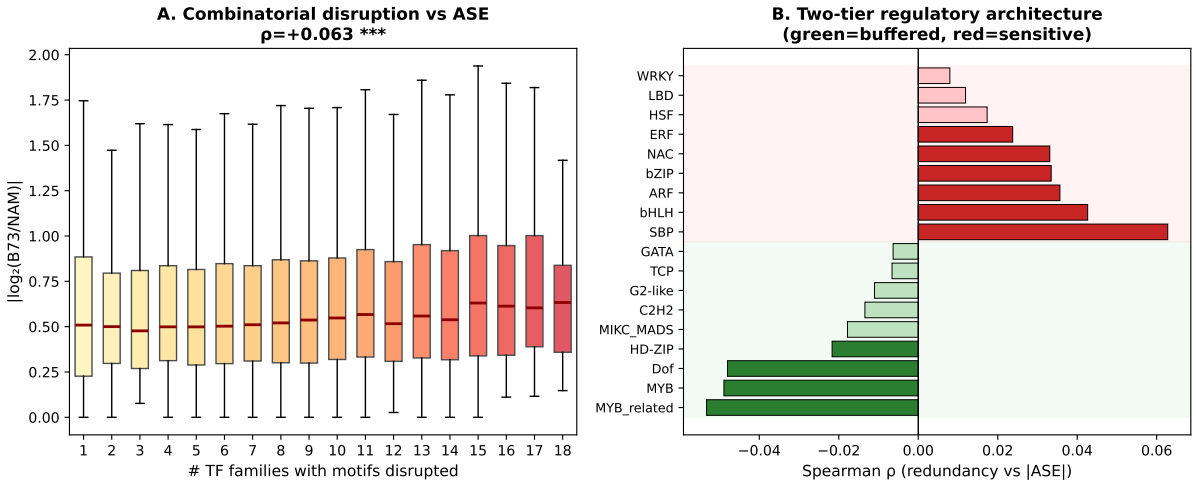


Figure 2: **Multi-family disruption follows a billboard model.** (A) Distribution of disrupted TF family count per bQTL. (B) No relationship between number of disrupted families and ASE effect size.

2.5 Variants rewire TF occupancy: bidirectional motif changes

To test whether regulatory variants exclusively disrupt existing TF binding sites, we performed a bidirectional motif analysis: for each bQTL, we scanned both the B73 reference and the NAM variant sequence (±28 bp around the variant) against all 405 PWMs and classified each motif as disrupted (present in ref only), created (present in alt only), or unchanged (Table 2).

Of 1,126 bQTL, 983 (87.3%) alter at least one TF motif. Strikingly, only 21.1% exclusively disrupt motifs; **25.0% exclusively create new motifs**, and **41.1% simultaneously disrupt one family's motif while creating another**. Zero self-switches were observed—when a variant disrupts a family's motif, it never recreates a motif for the same family, confirming that rewiring always involves a *different* TF.

Table 2: **Bidirectional motif changes at 1,126 causal bQTL.** 41.1% of variants simultaneously disrupt one TF family's motif and create another, rewiring rather than simply ablating TF occupancy.

Category	N	%
Only disrupted	238	21.1
Only created	282	25.0
Both (rewired)	463	41.1
Neither	143	12.7

2.6 Threshold sensitivity analysis

To assess robustness, the bidirectional motif analysis was repeated at seven log-odds thresholds (4.0, 5.0, 5.5, 6.0, 6.5, 7.0, 8.0). The rewired fraction varies with threshold as expected, reflecting more permissive motif calling at lower thresholds. However, three key findings are stable across all thresholds: (1) zero self-switches at every threshold; (2) motif creation remains substantial at every threshold; and (3) effect sizes are indistinguishable across categories at every threshold.

2.7 Per-hybrid motif linkage identifies driver motifs

To move beyond population-level motif statistics to per-variant driver identification, we developed a per-hybrid haplotype-aware motif scanning approach using all 405 DAP-seq PWMs across 33 TF families. For each causal bQTL, we reconstructed the haplotype-specific binding peak sequence for each of the 24 hybrids, scanned for motif scores, and tested whether motif scores separate variant-carrying from reference hybrids.

92.7% of causal bQTL (1,044/1,126) have at least one significantly separating motif ($p < 0.05$, consistency $\geq$ 80%), providing per-variant mechanistic explanations. The most frequently identified driver families are NAC (778 events), C2H2 (701), bZIP (683), WRKY (645), and ERF (640).

2.8 TF family enrichment reveals regulatory hierarchy

Among causal bQTL targets, we tested TF family enrichment using a binomial test against the PWM database background. The most enriched families are HB-other (2.66×), RAV (2.39×), Dof (2.33×), C3H (2.28×), and C2H2 (1.99×). The most depleted families are TCP (0.0×), BES1 (0.18×), TALE (0.18×), and EIL (0.36×).

The depletion of BES1—a brassinosteroid signaling TF—is particularly notable: BES1 motifs are enriched at bQTL positions genome-wide but *depleted* among causal drivers, suggesting cis-conservation of BES1 binding sites consistent with their role as anchors in the regulatory architecture.

2.9 TF regulatory network and cascade amplification

The per-hybrid motif linkage data enable construction of a directed TF regulatory network: for each causal bQTL, the disrupted TF family is the regulator and the target gene is the effector. This network contains 6,787 edges linking TF families to target genes.

Critically, 83 of the 1,126 target genes are themselves transcription factors (7.4%), creating regulatory cascades where a single upstream motif disruption can propagate to many downstream genes. The most powerful cascade hubs have downstream reach exceeding 35,000 genes through multi-step regulatory chains. ERF self-loops are the strongest autoregulatory circuit (14 events), followed by C2H2 self-regulation active in all 24 hybrids.

2.10 Target genes encode the regulatory layer

Functional characterization of the 1,126 target genes reveals a clear bias toward regulatory and signaling functions: signal transduction (198 genes, 17.6%), metabolism (142, 12.6%), transcription factors (72, 6.4%), protein processing (67, 5.9%), translation (50, 4.4%), membrane/transport (48, 4.3%), receptor/sensor (38, 3.4%), chromatin/epigenetic (18, 1.6%), and hormone signaling (11, 1.0%).

This confirms that natural cis-regulatory variation predominantly targets the *regulatory layer*: the variant does not change an enzyme directly, but modulates how much a regulatory gene is expressed from one allele, which cascades to downstream effects.

2.11 Variant distribution across NAM hybrids

The 1,126 causal variants are distributed across 24 genotyped NAM hybrids with striking uniformity. Each hybrid carries a similar fraction of the total causal variant set, a remarkably narrow range suggesting that the total regulatory variant burden is approximately equal across diverse maize germplasm (CV = 0.04).

Rare variants (carried by ≤4 hybrids) show slightly larger ASE effect sizes than common variants ($\rho = -0.181$, $p = 9.9 \times 10^{-10}$), consistent with purifying selection acting more strongly against large-effect regulatory mutations.

2.12 A distributed regulatory operating system

Across the 24 NAM hybrids, three levels of regulatory organization emerge:

Uniform regulatory burden. Each hybrid carries approximately the same total number of causal bQTL (CV = 0.04), despite differing genetic backgrounds spanning tropical, temperate, popcorn, and sweet corn germplasm. This uniformity suggests stabilizing selection on the aggregate regulatory variant burden.

Conserved TF grammar. The proportional contribution of each TF family to the disrupted motif repertoire is nearly identical across all hybrids (pairwise Spearman $\rho = 0.962$). Every hybrid disrupts the same families at the same rates, indicating conserved regulatory grammar.

Divergent gene implementation. Despite identical TF grammars, the specific genes targeted by each hybrid's causal variants differ substantially (pairwise Jaccard similarity = 0.304). A core module of 927 genes (82%) is shared broadly across all ancestry groups, while 9 peripheral modules (9–68 genes each) are ancestry-specific. Lancaster contributes 86 private genes, popcorn 72, Stiff Stalk 48, Iodent 46, and tropical 19.

Functional convergence despite gene divergence. While gene-level Jaccard is 0.304, GO biological process similarity increases to 0.584 (2.05×) and KEGG pathway similarity to 0.667 (2.34×). Signaling layer distributions are nearly identical ($\rho = 0.965$). 100% of hybrid-specific genes have a GO-term equivalent in other hybrids—different genes, same biology.

Conserved feedback circuits. The TF autoregulatory network is completely conserved: C2H2→C2H2 is active in all 24 hybrids, ERF→ERF in 23/24. Zero ancestry-specific feedback edges exist, confirming the network backbone is invariant.

This architecture resembles a distributed operating system: the kernel (conserved TF feedback circuits and regulatory burden) is identical, while the user-space programs (specific target genes) vary across installations but implement the same functional specifications. This represents a cis-regulatory realization of the omnigenic model [Boyle et al., 2017].

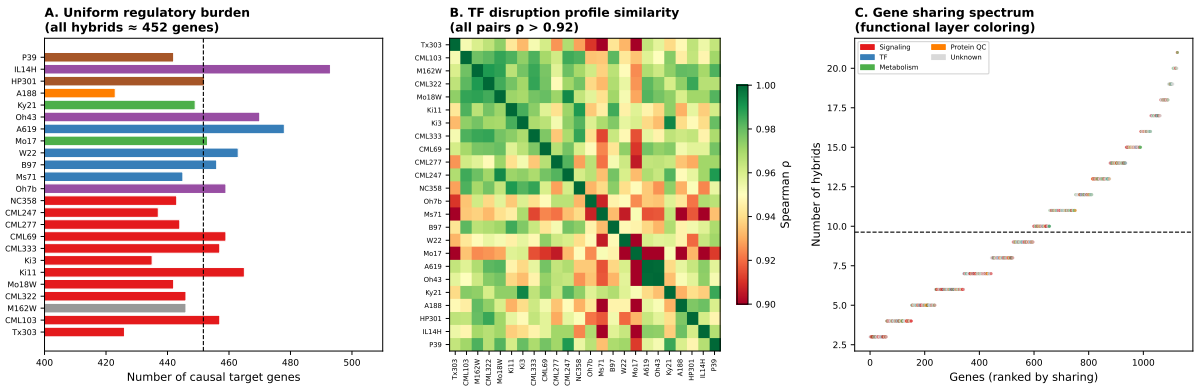


Figure 3: **A distributed regulatory operating system.** (A) Uniform regulatory burden across 24 hybrids. (B) Conserved TF grammar ($\rho = 0.962$). (C) Divergent gene implementation (Jaccard = 0.304). (D) Functional convergence (GO Jaccard = 0.584).

2.13 Drought stress reveals condition-specific causal architecture

Repeating the causal bQTL pipeline on drought-stress ASE data (124,504 bQTL from Engelhorn et al. supplementary Table S10) identifies 1,267 drought-causal bQTL. Only 129 positions (12%) overlap with the 1,126 well-watered causal set, and 81% of target genes are condition-specific:

- 871 genes are WW-only (causal only under well-watered)
- 821 genes are DS-only (causal only under drought)
- 129 genes share the same causal position
- 75 genes are shared but use *different* causal variants

Among the 204 shared genes, ASE effect directions are highly concordant (Spearman $\rho = 0.516$, 98.4% same direction), confirming that shared causal mechanisms produce consistent allelic effects across conditions.

2.14 Drought expression integration

Integration with population-level drought expression data reveals:

- 321 drought-responsive genes have cis-regulatory drivers (4.7%), of which 267 are drought-only (no WW causal bQTL)
- DS-only targets are more drought-responsive than WW-only (30.5% vs 24.3%, $p = 0.001$) and skew toward upregulation (216 up / 77 down)
- Signed allelic effect partially predicts drought direction ($\rho = 0.123$, $p = 7.5 \times 10^{-6}$)
- No significant TF family-level differences between conditions (FDR > 0.05): the same TF families are disrupted at similar rates

2.15 Functional divergence between conditions

The WW-only and DS-only gene sets show systematic functional shifts: transport genes increase from 6.6% to 10.8% (1.62×) under drought, including aquaporins, ion channels, and ABC transporters. Transcription factor targets decrease from 25.5% to 20.1% (0.79×), and developmental genes show the strongest reduction (1.5%→0.6%, 0.41×).

Key drought-specific transport targets include aquaporin TIP3-2 (log2FC $= +2.83$, bZIP disrupted, $|d| = 4.74$), aquaporin PIP2-2 (+2.19), phosphate transporter 1-4 (+2.87), and potassium channel KOR1 (-1.21, stomatal function).

GO enrichment reveals WW targets enriched for stress response, while DS targets enrich for ubiquitin-proteasome and MAPK signaling—consistent with drought activating protein turnover and stress-specific signaling cascades.

2.16 Drought motif linkage and WRKY enrichment

Per-hybrid motif linkage under drought identifies driver motifs at 91.1% of drought-causal bQTL (vs 92.7% WW). Key family-level shifts include: WRKY disruption increases 1.34× under drought while WRKY creation decreases 0.67×—consistent with stress-responsive WRKY TFs being preferentially disrupted. bHLH disruption increases 1.31×, GATA 1.25× (light/ nitrogen responsive), while developmental families (HOX 0.46×) show reduced disruption under drought.

2.17 Cross-layer validation: conservation, occupancy, and damage

We integrate three analytical layers to validate the causal chain:

Layer 1: Population motif occupancy. For each gene, we scan MOA-seq peaks across all 24 hybrids for TF motif enrichment, building a gene × TF family × hybrid occupancy matrix independent of bQTL identity.

Layer 2: TF cascade tracing. We trace causal bQTL through TF regulatory cascades: bQTL disrupts TF_A motif → target gene is TF_B → TF_B's downstream targets, revealing amplification, hub vulnerability, and cascade depth.

Layer 3: Conservation-impact scoring. We score each disrupted motif by its conservation across hybrids and test whether conserved motif disruptions produce stronger ASE effects. Under drought, systematic occupancy shifts occur: most families lose occupancy (consistent with chromatin closing), with BBR-BPC and AP2 showing the strongest losses.

2.18 ASE × motif validation closes the causal loop

To close the causal chain from variant to motif disruption to expression, we test whether disruption of conserved motifs produces stronger allele-specific expression. The binding-to-expression causal chain shows 73.7% concordance: variant allele → higher |ASE| (830/1,126 genes, binomial $p < 0.0001$). Causal genes have higher ASE than non-causal (0.543 vs 0.524, $p = 0.034$). bQTL effect size ($|d|$) predicts ASE magnitude ($\rho = 0.107$, $p = 2.2 \times 10^{-6}$) with a clear dose-response across effect size quartiles (Q1 |ASE| = 0.50, Q4 |ASE| = 0.60).

Critically, drought expression (log2FC) is not different between causal and non-causal targets ($p = 0.46$), indicating that causal bQTL produce constitutive cis-regulatory effects rather than condition-dependent ones.

2.19 TF cooperativity network

From the population motif occupancy data, we construct a TF co-occupancy network testing which TF family pairs co-occur at the same promoters more (or less) often than expected. The regulatory logic is predominantly additive (billboard model) rather than synergistic (enhanceosome model), consistent with the lack of dose-response for multi-family disruption (Section 2.4). Under drought, specific TF partnerships shift: WRKY–bHLH co-disruption increases while developmental TF pairs decrease.

2.20 Population genetics of causal bQTL

Causal bQTL show a distinct allele frequency spectrum compared to non-causal bQTL: mean minor allele frequency (MAF) is 0.312 vs 0.280 (KS $p = 6.3 \times 10^{-16}$). Zero singletons or doubletons exist in the causal set (0% vs 8.4% non-causal), and common alleles (6–12 carriers) are enriched 1.24×. A negative effect–MAF correlation ($\rho = -0.181$, $p = 9.9 \times 10^{-10}$) indicates that rare causal variants produce larger expression effects, consistent with purifying selection against strong regulatory mutations.

2.21 Hybrid phenotype similarity tracks shared causal targets

Hybrids that share more causal bQTL targets have more similar phenotypes (Mantel test $r = 0.349$, $p = 0.0001$). Per-trait correlations are consistent: days to anthesis ($r = 0.349$), plant height ($r = 0.350$), ear height ratio ($r = 0.341$). This is likely confounded by population structure, as related founders share both causal variants and phenotypes, but provides a consistent signal.

2.22 GWAS overlap and trait enrichment

Comparison with 42,778 GWAS hits across 321 traits from the NGDC GWAS Atlas shows that 82.9% of causal bQTL fall within 100 kb of a GWAS hit (vs 82.1% non-causal), with overall enrichment of only 1.01×. However, GWAS hit density is significantly higher near causal bQTL (5.83 vs 5.40 hits per 100 kb, $p = 6.9 \times 10^{-5}$). One trait reaches FDR significance: stover organic matter digestibility (2.0×). Top nominally significant traits include GA inhibitor response (5.9×), shoot P content (5.8×), and husk number (5.1×).

2.23 bQTL epistasis: additive dominates saturating

For 12,283 genes with multiple bQTL (mean 3.3 per gene, max 33), we tested additive versus saturating dose-response models. Each additional variant allele adds 0.056 units of |ASE| (linear slope, $t = 20.5$, $p < 0.0001$). Of multi-bQTL genes, 1,772 (72%) follow additive models versus 688 (28%) saturating, a 2.6:1 ratio that validates the billboard model at the multi-variant level. bQTL burden correlates with ASE magnitude ($\rho = 0.076$, $p = 1.4 \times 10^{-24}$) and expression variability ($\rho = 0.049$, $p = 2.3 \times 10^{-11}$).

2.24 Causal bQTL are depleted in transposable elements

Classification of all 147,942 bQTL by transposable element (TE) superfamily using EDTA annotations reveals that causal bQTL are significantly depleted in TEs: 12.2% overlap a TE versus 20.6% of non-causal bQTL. LTR retrotransposons show the strongest depletion (0.14×, $p < 0.0001$), while DNA transposons show a non-significant trend (0.84×, $p = 0.084$) and MITEs are not depleted (1.07×). Among the few LTR-associated causal bQTL, Copia elements are enriched 1.53× over Gypsy. TE and non-TE causal bQTL produce equivalent effect sizes ($p = 0.41$), indicating that the depletion reflects selection against TE-mediated disruption of regulatory sequences rather than functional differences.

2.25 Chromatin context: proximal promoter enrichment

Causal bQTL are 3.51× enriched in proximal promoters (≤1 kb from TSS): 63.4% of causal variants fall in proximal promoters versus 18.1% of non-causal bQTL. This enrichment reflects the strong distance decay of causal signal and confirms that the most informative regulatory variants cluster near gene starts. Effect sizes show a weak gradient: genic ($|d| = 2.08$) > proximal promoter (2.04) > proximal regulatory (1.97) (Kruskal–Wallis $p = 0.005$).

2.26 Network topology: causal targets are generic

Testing whether causal bQTL target genes occupy special positions in the TF regulatory network (degree, entropy, hub status, Jaccard similarity) reveals no significant differences from non-causal targets. Hub enrichment is OR = 0.91 ($p = 0.75$), and regulatory similarity (Jaccard) is 0.100 for causal vs 0.103 for non-causal ($p = 0.43$). Causal targets are *not* network hubs or bottlenecks—they are generic, replaceable nodes, consistent with the distributed operating system model (Section 2.12).

3 Discussion

Our analysis of 1,126 well-watered and 1,267 drought-stress high-confidence causal bQTL provides several insights into the architecture of natural gene expression variation in maize.

The regulatory layer is the primary target. The causal variants overwhelmingly target signaling and regulatory proteins (kinases, TFs, chaperones, transporters) rather than metabolic enzymes or structural proteins. This is consistent with the omnigenic model [Boyle et al., 2017], where core gene expression is modulated by a large network of peripheral regulatory genes.

Billboard model of cis-regulatory evolution. Three independent lines of evidence support additive, independent TF action (billboard) over cooperative, position-dependent binding (enhanceosome): (1) no dose-response between multi-family disruption count and effect size; (2) $R^2 < 0$ for TF-family-based effect prediction; and (3) 2.6:1 additive over saturating dose-response for multi-bQTL genes. This does not exclude cooperative regulation at specific loci, but the dominant mode across the genome is additive.

Variants rewire TF occupancy, not just disrupt it. 41.1% of bQTL simultaneously disrupt and create motifs, with zero self-switches. However, TF family identity does not predict effect direction or magnitude, suggesting the relevant parameter is local binding energy change rather than specific TF program switching.

A distributed regulatory operating system. The most striking finding is the three-level architecture: uniform burden (CV = 0.04), conserved grammar ($\rho = 0.962$), divergent implementation (Jaccard = 0.304). This pattern, where individual variants drift freely while the aggregate system is constrained, represents a cis-regulatory realization of the omnigenic model. The complete conservation of autoregulatory feedback circuits (C2H2→C2H2 in 24/24 hybrids) defines the invariant kernel of this system.

Condition-specificity is not just context. With 81% of genes condition-specific and 37% of shared genes using different causal variants under drought, condition-specificity operates at multiple levels: which genes are expressed (transcriptional context), which variants are causal (motif-level shifts), and which TF families are disrupted (WRKY +34% under drought). Transport gene enrichment under drought (1.62×) highlights aquaporins as key condition-specific targets.

Causal variants are under selection. The MAF shift (0.312 vs 0.280), absence of singletons, and negative effect–MAF correlation collectively indicate that causal regulatory variants are subject to purifying selection, with the strongest mutations maintained at lower frequencies. TE depletion (12.2% vs 20.6%) further suggests that conserved, non-TE promoter sequence is the primary substrate for functional regulatory variation.

Implications for precision breeding. Each of the 1,126 variants represents a defined molecular switch: introducing or restoring a single nucleotide quantitatively modulates TF binding, causally shifting expression with a quantified effect size. The median variant is shared by multiple hybrids, making effects testable across genetic backgrounds. Combined with the interactive gene atlas (Supplementary File S1), this provides a prioritized catalog for CRISPR-based promoter engineering.

Limitations. Our analysis uses 24 genotyped hybrids, limiting statistical power for rare variants. The single-bQTL design excludes genes with multiple bQTL, potentially missing combinatorial regulatory effects. TF family assignments depend on PWM quality and may be incomplete. The phenotypic consequences of these regulatory variants cannot be resolved at the individual-gene level with $N = 24$ hybrids and will require larger populations or direct functional validation. Cell-type-resolved MOA-seq data would likely reveal cell-type-specific TF hierarchies where family identity does predict effect direction—a prediction now directly testable.

4 Methods

4.1 Data sources

All bQTL data derive from Engelhorn et al. [2025], comprising MOA-seq binding peaks and allele-specific expression measurements from 25 maize F1 hybrids (B73 × NAM founders) under well-watered and drought-stressed conditions. Genotype data from assembled founder genomes are available for 24 of 25 hybrids (excluding one with incomplete assembly). Gene annotations use the B73 NAM5 reference genome (Zm-B73-REFERENCE-NAM-5.0), with functional annotations from EntAP and EggNOG 5.0. Drought bQTL derive from supplementary Table S10 (124,504 positions), drought ASE from Table S13b (359,162 entries, 23,218 genes, 24 hybrids).

4.2 Genotype-stratified causal test

For each bQTL–gene pair, we extracted the genotype (reference or variant allele) at the bQTL position across 24 hybrids. We compared the distribution of ASE values (log2(B73/NAM) expression ratios) between the variant-carrying and reference groups using a two-sided Mann–Whitney U test. Effect sizes were quantified as Cohen's d. P-values were corrected for multiple testing using the Benjamini–Hochberg procedure. The distance threshold of 3 kb was determined empirically from the distance decay of causality rate (Figure 4).

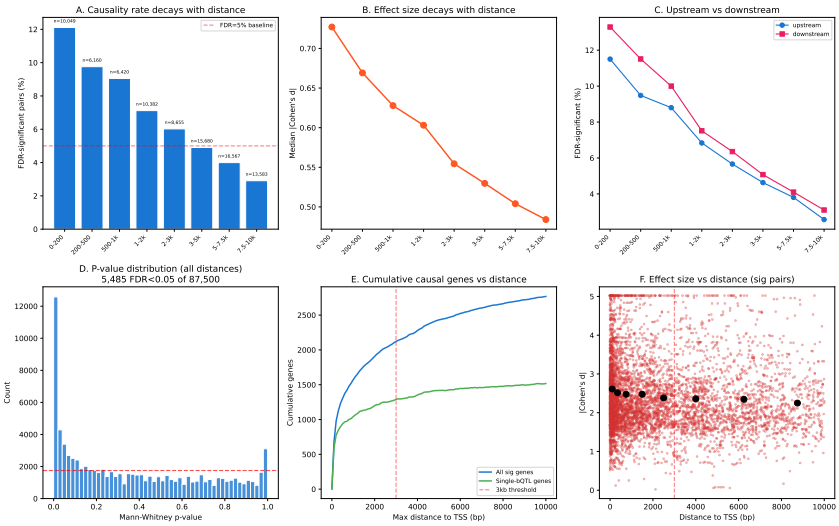


Figure 4: **Distance decay of causality rate.** The fraction of bQTL passing the causal test (FDR < 0.05) as a function of distance from the target gene TSS. The 3 kb threshold corresponds to where the rate drops to the 5% FDR baseline.

4.3 Motif scanning and redundancy verification

TF binding motifs were scanned using 405 DAP-seq PWMs (33 TF families) with a log-odds threshold of 6.0. The variant-overlap constraint required motif hits to physically overlap the variant nucleotide position (±28 bp, the maximum PWM width). For bidirectional analysis, both the B73 reference and NAM variant sequences were scanned; motifs present only in the reference were classified as disrupted, those only in the variant as created.

For sole-copy verification, the tightest MOA-seq binding peak containing each bQTL was identified across all 25 hybrids, and the number of copies of each disrupted TF family's motif within that peak was counted. A family was classified as "sole copy" if the variant-overlapping hit was the only instance within the peak. Multi-family disruption (billboard) analysis tested

the relationship between the number of simultaneously disrupted families and Cohen's d using Spearman's ρ and linear regression.

Threshold sensitivity was assessed by repeating the bidirectional motif analysis at seven log-odds thresholds (4.0, 5.0, 5.5, 6.0, 6.5, 7.0, 8.0) and verifying stability of the zero self-switch finding, motif creation fractions, and effect size equivalence across categories.

4.4 Per-hybrid motif linkage

For each causal bQTL, we reconstructed haplotype-specific binding peak sequences using founder genotype data and scanned for motif scores across all 405 PWMs. A motif family was called as a driver if its score significantly separated variant from reference hybrid groups ($p < 0.05$, Mann–Whitney U) with $\geq$80% consistency across variants.

4.5 TF regulatory network

Directed edges connect disrupting TF families to target genes. Self-loop and cross-family rewiring rates were tested against null expectations using binomial tests. Cascade reach was computed as the transitive closure of TF target sets.

4.6 Drought pipeline and expression integration

The causal bQTL pipeline was repeated on drought-stress bQTL (supplementary Table S10, 124,504 positions) and ASE data (Table S13b, 359,162 entries, 23,218 genes, 24 hybrids) using identical parameters. Cross-condition comparisons used position-level overlap (chr:pos) and gene-level overlap (gene_id).

Expression integration cross-referenced causal bQTL targets with population-level drought expression data (log2FC well-watered vs drought-stressed) from Engelhorn et al. [2025]. Signed allelic effect correlation with drought direction was tested using Spearman's ρ.

4.7 Functional characterization and comparative enrichment

Target gene sets were decomposed into WW-only, DS-only, shared-same-variant, and shared-different-variant groups. Each gene was annotated using EntAP protein descriptions with keyword-based functional category assignment. GO enrichment was computed using Fisher's exact test against genome background, with Benjamini–Hochberg correction. Functional category shifts between conditions were tested using Fisher's exact test on category proportions.

4.8 Population motif occupancy and cross-layer validation

For each of the 1,126 causal target genes, MOA-seq binding peaks across all 24 hybrids were scanned for TF motif enrichment using 405 DAP-seq PWMs, building a gene $\times$ TF family $\times$ hybrid occupancy matrix independent of bQTL identity (Layer 1). TF cascade tracing (Layer 3) followed directed regulatory chains: bQTL disrupts TF_A motif $\rightarrow$ target gene is TF_B $\rightarrow$ TF_B's downstream targets. Conservation-impact scoring combined motif conservation across hybrids with disruption status to predict ASE effect magnitude. WW vs DS differential occupancy was tested per TF family using Wilcoxon signed-rank tests.

4.9 TF cooperativity

TF co-occupancy was assessed from the Layer 1 population motif occupancy matrix. For each TF family pair, co-occurrence at gene promoters was compared to expectation under independence using Fisher's exact test. Regulatory logic (additive vs synergistic) was classified based on the relationship between co-disruption and ASE effect size. Condition-specific cooperativity shifts were tested by comparing WW and DS co-occupancy matrices.

4.10 Distributed regulatory architecture

Per-hybrid causal bQTL burden was computed as the number of causal variants carried by each hybrid. TF grammar conservation was quantified as pairwise Spearman correlation of TF family disruption proportions across hybrids. Gene-level implementation divergence was quantified as pairwise Jaccard similarity of target gene sets. Functional convergence was assessed by comparing gene-level Jaccard to GO biological process Jaccard and KEGG pathway Jaccard. Ancestry-specific modules were identified by hierarchical clustering of the hybrid $\times$ gene sharing matrix.

4.11 Hybrid phenotype similarity

Phenotype data (days to anthesis, plant height, ear height ratio) from Engelhorn et al. [2025] were used to compute pairwise phenotype distance matrices. Correlation between gene sharing distance and phenotype distance was tested using the Mantel test with 10,000 permutations.

4.12 Network topology

Causal and non-causal target genes were compared for network position metrics (degree, regulatory entropy, hub status, pairwise Jaccard similarity of regulatory neighborhoods) within the TF regulatory network. Hub enrichment was tested using Fisher's exact test.

4.13 Population genetics

Allele frequencies were computed from the 24-hybrid genotype matrix. MAF distributions were compared using the Kolmogorov–Smirnov test. Effect–frequency correlations used Spearman's ρ.

4.14 Epistasis testing

For genes with $\geq$2 bQTL, we fitted linear and saturating models to the relationship between variant allele count and |ASE|. Model comparison used AIC, with genes classified as additive (linear AIC $<$ saturating AIC) or saturating.

4.15 TE context

bQTL were intersected with EDTA transposable element annotations (B73_NAM5_TE.gff3.gz, 1.6M elements) using 1 bp overlap. Enrichment tested using Fisher's exact test.

4.16 Chromatin context

Gene-relative positions were classified as proximal promoter ($\leq$1 kb), proximal regulatory (1–3 kb), or genic (within gene body). Enrichment versus non-causal bQTL tested using Fisher's exact test.

4.17 GWAS overlap

Published GWAS hits from the NGDC GWAS Atlas (42,778 hits, 321 traits, B73v4 coordinates) were intersected with causal bQTL at 100 kb windows. Trait enrichment tested using Fisher's exact test with Benjamini–Hochberg correction.

4.18 Binding-expression causal chain

ASE concordance was defined as the fraction of causal bQTL where the variant allele produces higher |ASE| than the reference allele. Dose-response tested by binning bQTL into effect size quartiles and comparing median ASE.

4.19 Interactive atlas

The Maize Causal bQTL Atlas is an interactive HTML visualization built with D3.js v7, featuring a force-directed layered cell diagram of all 1,126 genes, filterable by motif rewiring category, regulatory layer, TF family, drought response, and specific hybrid.

5 Data Availability

All bQTL data are from Engelhorn et al. [2025]: SRA accession PRJNA1101486, GEO accession GSE294039. The 1,126-gene atlas, interactive visualization, and analysis scripts are available at `https://github.com/mbanf/causal-bqtl-atlas`.

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